

Assignment of linear algebra

Unit 1: Linear Equation and Matrices

Group A

1. Define system of linear equation with example with three variables.
2. What is the Row echelon form of the matrix and Row reduced echelon form of the matrix with suitable examples?
3. Define vector equation.
4. Define linear combination, Linear dependence, and linear independence.
5. When the system of linear equation consistence or inconsistency?
6. Illustrate by an example that a system of linear equation has either exactly one solution or infinitely many solution.
7. Define singular and non-singular matrix.

Group B

1. Let $A = \begin{bmatrix} 2 & 5 \\ -3 & -7 \end{bmatrix}$ and $B = \begin{bmatrix} -7 & -5 \\ 3 & 2 \end{bmatrix}$

Show that $AB = BA = I$

2. Let $A = \begin{bmatrix} 2 & 3 & 1 \\ -3 & -7 & 0 \\ 4 & 5 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} -7 & 5 & 0 \\ 3 & 2 & 4 \\ -1 & 2 & 7 \end{bmatrix}$

Find AB , BA , A^2 , B^2 .

3. Reduce the augmented matrix of the system in reduced echelon form of the following matrices.

a. $\begin{bmatrix} 0 & 1 & 0 & -3 & 3 \\ 1 & 0 & 3 & 0 & 2 \\ 0 & -2 & 3 & 2 & 1 \\ 3 & 0 & 0 & 7 & -5 \end{bmatrix}$

b. $\begin{bmatrix} 0 & -3 & -6 & 4 & 9 \\ -1 & -2 & -1 & 3 & 1 \\ -2 & -3 & 0 & 3 & -1 \\ 1 & 4 & 5 & -9 & -7 \end{bmatrix}$

4. Given the matrix $\begin{bmatrix} 0 & 3 & 6 & 6 & -5 \\ 3 & -7 & 8 & -5 & 9 \\ 3 & -9 & 12 & -9 & 15 \end{bmatrix}$, discuss

the forward phase and backward and backward of row reduction algorithm.

5. Solve the following equation

a. $2x+3y+4z=20$	b. $x_2-8x_3=8$	c. $2x_1-5x_2+8x_3=0$
$3x+4y+5z=26$	$2x_1-3x_2+2x_3=1$	$-2x_1-7x_2+x_3=0$
$3x+5y+6z=31$	$5x_1-8x_2+7x_3=1$	$4x_1+2x_2+7x_3=0$

Unit 2. Matrix Algebra

Group A

1. List any two characteristics of invertible matrices.
2. Define the Partitioned matrices.
3. Write the numerical importance of partitioning matrices?
4. Describe the Leontief input-output closed model, input output open model, technical matrices, and Leontief matrices. What are the viable conditions of Leontief matrix?

Group B

1. Find the inverse matrix of the following by using elementary row reduced augmented matrix.

a. $A = \begin{bmatrix} 1 & 2 \\ 7 & -3 \end{bmatrix}$

b. $A = \begin{bmatrix} 1 & -3 & 2 \\ -3 & 3 & -1 \\ 2 & -1 & 0 \end{bmatrix}$

c. $A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & 3 \\ 4 & -3 & 8 \end{bmatrix}$

2. Compute the multiplication of Partitioned matrices for

$$A = \begin{bmatrix} 2 & -3 & 1 & 0 & -4 \\ 1 & 5 & -2 & 3 & -1 \\ 0 & 4 & -2 & 7 & -1 \end{bmatrix}$$

$$B = \begin{bmatrix} 6 & 4 \\ -2 & 1 \\ -3 & 7 \\ -1 & 3 \\ 5 & 2 \end{bmatrix}$$

3. Consider Leontief input-output model equation $X = AX + d$, where Consumption matrix,

$$A = \begin{bmatrix} 0.50 & 0.40 & 0.20 \\ 0.20 & 0.30 & 0.10 \\ 0.10 & 0.10 & 0.30 \end{bmatrix}. \text{ Suppose final demand is}$$

50 units manufacturing, 30 units for agriculture and 20 units for service. Find the production level X that will satisfy the demand. (Page 11 example 18)

Unit 6. Eigen values and Eigen vectors.

Group A

1. Define eigen values.
2. Define eigen vectors.
3. Define the characteristic equation.
4. Define the characteristic matrix.
5. Define the characteristic polynomial.
6. Define Complex eigen value.
7. State Caley's Hamilton Theorem.

Group B

1. a. Is $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$ an eigen vector of $\begin{pmatrix} 1 & 4 \\ 3 & 2 \end{pmatrix}$?

b. Is $\begin{pmatrix} 3 \\ -2 \end{pmatrix}$ an eigen vector of $\begin{pmatrix} 1 & 6 \\ 5 & 2 \end{pmatrix}$?

c. Is $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$ an eigen vector of $\begin{pmatrix} 3 & 0 \\ 8 & -1 \end{pmatrix}$?

d. Is $\begin{pmatrix} 6 \\ -5 \end{pmatrix}$ an eigen vector of $\begin{pmatrix} 1 & 6 \\ 5 & 2 \end{pmatrix}$?

e. Is $\begin{pmatrix} 4 \\ -3 \\ 1 \end{pmatrix}$ an eigen vector of $\begin{pmatrix} 3 & 7 & 9 \\ -4 & -5 & 1 \\ 2 & 4 & 4 \end{pmatrix}$?

2. a. Is $\lambda = 1$ an eigenvalue of matrix $\begin{pmatrix} 2 & 2 \\ 3 & 5 \end{pmatrix}$?

b. Is $\lambda = 5$ an eigenvalue of matrix $\begin{pmatrix} 5 & 0 \\ 2 & 1 \end{pmatrix}$?

c. Is $\lambda = 7$ an eigenvalue of matrix $\begin{pmatrix} 1 & 6 \\ 5 & 2 \end{pmatrix}$?

d. Is $\lambda = 2$ an eigenvalue of matrix $\begin{pmatrix} 4 & -1 & 6 \\ 2 & 1 & 6 \\ 2 & -1 & 8 \end{pmatrix}$?

3. Find the eigen value and eigen vector of the following.

a. $\begin{pmatrix} 2 & 2 \\ 3 & 5 \end{pmatrix}$ b. $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ c. $\begin{pmatrix} 4 & 0 & 0 \\ 1 & 4 & 0 \\ 0 & 0 & 5 \end{pmatrix}$

4. Diagonalizable the following matrices

a. $\begin{pmatrix} 2 & 2 \\ 3 & 5 \end{pmatrix}$ b. $\begin{pmatrix} 4 & 0 & 0 \\ 1 & 4 & 0 \\ 0 & 0 & 5 \end{pmatrix}$

$$\text{c.} \begin{pmatrix} 2 & 4 & 3 \\ -4 & -6 & -3 \\ 3 & 3 & 1 \end{pmatrix}$$

$$\text{d.} \begin{pmatrix} 1 & 3 & 3 \\ -3 & -5 & -3 \\ 3 & 3 & 1 \end{pmatrix}$$

$$\text{e.} \begin{pmatrix} 5 & 0 & 0 & 0 \\ 0 & 5 & 0 & 0 \\ 1 & 4 & -3 & 0 \\ -1 & -2 & 0 & -3 \end{pmatrix}$$